

## Modular Arithmetic

Modular arithmetic is a special type of arithmetic that involves only integers.

### Notation:

$A \equiv B \pmod{C}$  is an abbreviation for  $A \pmod{C} = B \pmod{C}$

## Congruence Module

$$A \equiv B \pmod{C}$$

$A$  is congruent to  $B$  modulo  $C$

- $\equiv$  is the symbol for congruence, which means the values  $A$  and  $B$  are in the same equivalence class
- $\pmod{C}$  tells us what operation we applied to  $A$  and  $B$

### Example

$$26 \equiv 11 \pmod{5}$$

- $26 \pmod{5} = 1 \rightarrow$  it is in the equivalence class for 1
- $11 \pmod{5} = 1 \rightarrow$  it is in the equivalence class for 1 as well

In general, two integers  $a$  and  $b$  are congruent modulo  $n$  when  $a - b$  is a multiple of  $n$ .

In other words,  $a \equiv b \pmod{n}$  when  $(a-b)/n$  is an integer.

Otherwise,  $a \not\equiv b \pmod{n}$  which means that  $a$  and  $b$  are **not congruent** modulo  $n$ .

### Examples

- $43 \equiv 22 \pmod{7}$  because  $\frac{43 - 22}{7} = \frac{21}{7} = 3$  which is an integer

## Summary of Useful Facts

Consider four integers  $a, b, c, d$  and a positive integer  $m$  such that:

$$a \equiv b \pmod{m}$$

$$c \equiv d \pmod{m}$$

In modular arithmetic, the following identities hold:

- Addition:

$$a + c \equiv b + d \pmod{m}$$

- Subtraction:

$$a - c \equiv b - d \pmod{m}$$

- Multiplication:

$$ac \equiv bd \pmod{m}$$

- Division:

$$\frac{a}{e} \equiv \frac{b}{e} \pmod{\frac{m}{\gcd(m, e)}}$$

where  $e$  is a positive integer that divides  $a$  and  $b$ .

- Exponentiation:

$$a^e \equiv b^e \pmod{m}$$

where  $e$  is a positive integer.